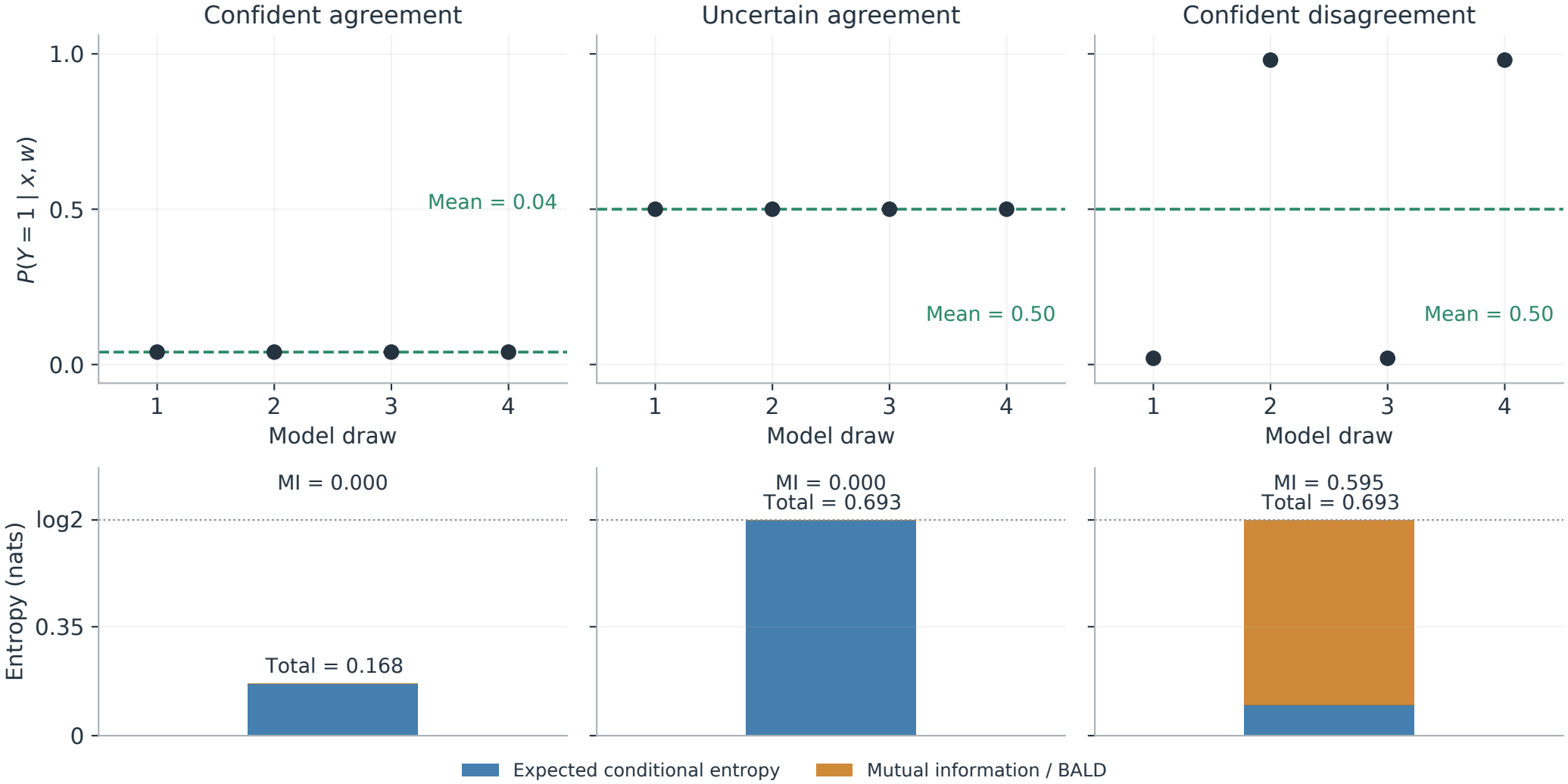


$H(\overline{p}) = \mathbb{E}_q H(p_w) + I_q(Y; W \mid x)$: the last two cases have identical average predictions.

This is a decomposition under the chosen model distribution q, not an automatic measurement of true data noise.